

Factorized double-commutator equations

This note extracts the equations implemented in `src/ReferenceImplementations.cc` and `src/FactorizedDoubleCommutator.cc` for

$$Z = [\eta, [\eta, \Gamma]_{3b}]_{1b, 2b}, \quad (1)$$

where all two-body matrix elements are the non-reduced J-coupled matrix elements used by the code.

Notation

$$\bar{n}_a \equiv 1 - n_a, \quad d_a \equiv 2j_a + 1, \quad \hat{J}^2 \equiv 2J + 1. \quad (2)$$

The C++ code often stores `j2 = 2j`, so factors such as `op.j2 + 1.0` are written here as $d_p = 2j_p + 1$. Matrix elements are written as

$$\eta_{ab,cd}^J \equiv \langle ab; J | \eta | cd; J \rangle, \quad \Gamma_{ab,cd}^J \equiv \langle ab; J | \Gamma | cd; J \rangle. \quad (3)$$

The permutation symbols P_{pg} and P_{qh} mean that the code evaluates the direct term plus the exchanged terms with the corresponding stored two-body phase, e.g. `bra.Phase(J0)` or `ket.Phase(J0)`. The final matrix storage also divides by $\sqrt{2}$ when either pair contains identical orbits.

Reference implementation: `comm223_231_BruteForce`

Source: `src/ReferenceImplementations.cc`.

This routine builds the one-body part Z_{pq} directly by summing the full diagram expressions.

Diagram I

Source: `src/ReferenceImplementations.cc`.

$$Z_{pq}^I = \frac{1}{2} \frac{\delta_{j_p j_q}}{d_p} \sum_{abcde J_0 J_1} \delta_{j_d j_e} \frac{\hat{J}_0^2 \hat{J}_1^2}{d_d} \left(\bar{n}_a \bar{n}_c n_b n_d - \bar{n}_b \bar{n}_d n_a n_c - \bar{n}_b \bar{n}_e n_a n_c + \bar{n}_a \bar{n}_c n_b n_e \right) \\ \times \eta_{bd,ac}^{J_0} \eta_{ac,be}^{J_0} \Gamma_{ep,dq}^{J_1}. \quad (4)$$

Diagram IIa

Source: src/ReferenceImplementations.cc.

$$Z_{pq}^{IIa} = \frac{\delta_{j_p j_q}}{d_p} \sum_{abcde J_0 J_1 J_2 J_3} (-1)^{J_0+J_1+J_2+j_c+j_d} \hat{J}_0^2 \hat{J}_1^2 \hat{J}_2^2 \hat{J}_3^2 \left(\bar{n}_a \bar{n}_c n_b n_d - \bar{n}_b \bar{n}_d n_a n_c \right) \\ \times \begin{Bmatrix} j_a & j_b & J_3 \\ j_d & j_c & J_0 \end{Bmatrix} \begin{Bmatrix} j_p & j_e & J_3 \\ j_d & j_c & J_1 \end{Bmatrix} \begin{Bmatrix} j_b & j_p & J_2 \\ j_e & j_a & J_3 \end{Bmatrix} \eta_{bd,ac}^{J_0} \eta_{cp,de}^{J_1} \Gamma_{ae,bq}^{J_2}. \quad (5)$$

Diagram IIb

Source: src/ReferenceImplementations.cc.

$$Z_{pq}^{IIb} = \frac{1}{4} \frac{\delta_{j_p j_q}}{d_p} \sum_{abcde J_0} \hat{J}_0^2 \left(\bar{n}_a \bar{n}_d n_b n_e - \bar{n}_b \bar{n}_e n_a n_d \right) \eta_{be,ad}^{J_0} \eta_{cp,be}^{J_0} \Gamma_{ad,cq}^{J_0}. \quad (6)$$

Diagram IIc

Source: src/ReferenceImplementations.cc.

$$Z_{pq}^{IIc} = \frac{\delta_{j_p j_q}}{d_p} \sum_{abcde J_0 J_1 J_2 J_3} \hat{J}_0^2 \hat{J}_1^2 \hat{J}_2^2 \hat{J}_3^2 \left(\bar{n}_a \bar{n}_d n_b n_e - \bar{n}_b \bar{n}_e n_a n_d \right) \\ \times \begin{Bmatrix} j_d & j_e & J_3 \\ j_b & j_a & J_0 \end{Bmatrix} \begin{Bmatrix} j_c & j_p & J_3 \\ j_b & j_a & J_1 \end{Bmatrix} \begin{Bmatrix} j_e & j_c & J_2 \\ j_p & j_d & J_3 \end{Bmatrix} \eta_{be,da}^{J_0} \eta_{ca,bq}^{J_1} \Gamma_{dp,ce}^{J_2}. \quad (7)$$

Diagram IIId

Source: src/ReferenceImplementations.cc.

The code keeps this as a commented expression and notes that it is almost the same as IIb, so it is combined in the optimized implementation.

$$Z_{pq}^{IIId} = -\frac{1}{4} \frac{\delta_{j_p j_q}}{d_p} \sum_{abcde J_0} \hat{J}_0^2 \left(\bar{n}_c \bar{n}_d n_a n_e - \bar{n}_a \bar{n}_e n_c n_d \right) \eta_{ae,cd}^{J_0} \eta_{cd,bq}^{J_0} \Gamma_{bp,ae}^{J_0}. \quad (8)$$

Diagram IIIa

Source: src/ReferenceImplementations.cc.

$$Z_{pq}^{IIIa} = \frac{1}{2} \frac{\delta_{j_p j_q}}{d_p} \sum_{abcdeJ_0J_1} \delta_{j_d j_e} \frac{\hat{J}_0^2 \hat{J}_1^2}{d_d} \left(\bar{n}_a \bar{n}_e n_b n_c - \bar{n}_b \bar{n}_c n_a n_e \right) \eta_{bc,ae}^{J_0} \eta_{ep,dq}^{J_1} \Gamma_{ad,bc}^{J_0}. \quad (9)$$

Diagram IIIb

Source: src/ReferenceImplementations.cc.

$$Z_{pq}^{IIIb} = -\frac{1}{2} \frac{\delta_{j_p j_q}}{d_p} \sum_{abcdeJ_0J_1} \delta_{j_d j_e} \frac{\hat{J}_0^2 \hat{J}_1^2}{d_d} \left(\bar{n}_a \bar{n}_c n_b n_d - \bar{n}_b \bar{n}_d n_a n_c \right) \eta_{bd,ac}^{J_0} \eta_{ep,dq}^{J_1} \Gamma_{ac,be}^{J_0}. \quad (10)$$

Factorized implementation: comm223_231

Source: src/FactorizedDoubleCommutator.cc.

ArXiv correspondence: this section matches the one-body J-coupled factorized equations in learn/axiv.txt, namely

$$f^I, \quad f^{II}, \quad f^{III_a}, \quad f^{III_b} \quad (11)$$

in `doubleCommutator_1b_Jcoupled_factorized`, together with the intermediates

$$\chi^\alpha, \quad \chi^\beta, \quad \bar{\chi}^\gamma, \quad \chi^\delta \quad (12)$$

in `Jcoupled_chi_1b`.

The optimized one-body implementation splits the same result into a one-body-intermediate path and a two-body-intermediate path.

One-body intermediate, diagram I

Source: src/FactorizedDoubleCommutator.cc.

ArXiv correspondence tag: this is the code-backed version of

$$f_{ij}^I \quad (13)$$

with intermediate

$$\chi_{ij}^\alpha. \quad (14)$$

For a two-body channel with angular momentum J_0 , define the occupation-dressed matrix

$$\text{ArXiv correspondence: contributes to } \chi^\alpha \text{ in } \text{Jcoupled_chi_1b}. \quad (15)$$

$$\left(\eta_{nn\bar{n}\bar{n}}^{J_0}\right)_{ij,kl} = \left[n_i n_j \bar{n}_k \bar{n}_l - \bar{n}_i \bar{n}_j n_k n_l\right] \eta_{ij,kl}^{J_0}. \quad (16)$$

The code forms

$$\text{ArXiv correspondence: matrix-product realization of } \chi^\alpha. \quad (17)$$

$$T^{J_0} = 2\hat{J}_0^2 \eta_{nn\bar{n}\bar{n}}^{J_0} \eta^{J_0} + \left(2\hat{J}_0^2 \eta_{nn\bar{n}\bar{n}}^{J_0} \eta^{J_0}\right)^T, \quad (18)$$

and contracts it to

$$\text{ArXiv correspondence: } \chi_{ij}^\alpha. \quad (19)$$

$$\chi_{de}^{(221a)} = \frac{1}{d_d} \sum_{bJ_0} T_{bd,be}^{J_0}. \quad (20)$$

Then

$$\text{ArXiv correspondence: } f_{ij}^I \text{ in } \text{doubleCommutator_1b_Jcoupled_factorized}. \quad (21)$$

$$Z_{pq}^I = \frac{1}{2d_p} \sum_{deJ_1} \hat{J}_1^2 \chi_{de}^{(221a)} \Gamma_{ep,dq}^{J_1}. \quad (22)$$

One-body intermediate, diagrams IIIa and IIIb

Source: src/FactorizedDoubleCommutator.cc.

ArXiv correspondence tag: this is the code-backed version of

$$f_{ij}^\Pi \quad (23)$$

with intermediate

$$\chi_{ij}^\beta. \quad (24)$$

The non-Hermitian one-body intermediate is

$$\text{ArXiv correspondence: } \chi_{ij}^\beta \text{ in } \text{Jcoupled_chi_1b}. \quad (25)$$

$$\chi_{de}^{(221b)} = \frac{1}{d_d} \sum_{abcJ_0} \hat{J}_0^2 \left(\bar{n}_a \bar{n}_e n_b n_c - \bar{n}_b \bar{n}_c n_a n_e \right) \eta_{bc,ae}^{J_0} \Gamma_{ad,bc}^{J_0}. \quad (26)$$

The final contraction is

ArXiv correspondence: f_{ij}^{II} in `doubleCommutator_1b_Jcoupled_factorized`.
(27)

$$Z_{pq}^{IIa+IIb} = \frac{1}{2d_p} \sum_{deJ_1} \hat{J}_1^2 \left[\chi_{de}^{(221b)} - h_Z \chi_{ed}^{(221b)} \right] \eta_{ep,dq}^{J_1}. \quad (28)$$

Two-body intermediate, diagrams IIb and IIc

Source: `src/FactorizedDoubleCommutator.cc`.

ArXiv correspondence tag: this is the code-backed version of

$$f_{ij}^{\text{III}_b} \quad (29)$$

with intermediate

$$\chi_{ijkl}^\delta. \quad (30)$$

In each ordinary two-body channel, define

ArXiv correspondence: contributes to $\chi_{ijkl}^{\delta J}$ in `Jcoupled_chi_1b`.
(31)

$$\eta_{occ}^J(ij, kl) = \left[n_i n_j \bar{n}_k \bar{n}_l - \bar{n}_i \bar{n}_j n_k n_l \right] \eta_{ij,kl}^J. \quad (32)$$

The optimized code builds the matrix product

ArXiv correspondence: matrix-product realization of $\chi^{\delta J}$.
(33)

$$\chi^{(222b),J} = 4\hat{J}^2 \left(\eta^J \eta_{occ}^J \Gamma^J - \Gamma^J \eta_{occ}^J \eta^J \right), \quad (34)$$

with the equal-channel case using transpose symmetry. The one-body contraction is

ArXiv correspondence: $f_{ij}^{\text{III}_b}$ in `doubleCommutator_1b_Jcoupled_factorized`.
(35)

$$Z_{pq}^{IIb+IIc} = \frac{1}{4d_p} \sum_{cJ_0} \chi_{cp,cq}^{(222b),J_0}. \quad (36)$$

Pandya transform for diagrams IIa and IIc

Source: `src/FactorizedDoubleCommutator.cc`.

ArXiv correspondence tag: this is the code-backed version of

$$f_{ij}^{\text{III}_a} \quad (37)$$

with intermediate

$$\bar{\chi}_{ijkl}^{\gamma J}, \quad (38)$$

written in the Pandya/cross-coupled channel used by the code.

The code uses cross-coupled channels and the Pandya transform

$$\text{ArXiv correspondence: Pandya transform entering } \bar{\chi}^{\gamma J}. \quad (39)$$

$$\bar{X}_{ab,cd}^J = \sum_{J'} (-1)^{j_b+j_c+J'} \hat{J}'^2 \begin{Bmatrix} j_a & j_b & J \\ j_c & j_d & J' \end{Bmatrix} X_{ad,bc}^{J'}. \quad (40)$$

With

$$\text{ArXiv correspondence: occupation factor entering } \bar{\chi}^{\gamma J}. \quad (41)$$

$$f_{ab,cd} = \bar{n}_c \bar{n}_b n_a n_d - n_c n_b \bar{n}_a \bar{n}_d, \quad (42)$$

the intermediate matrix is

$$\text{ArXiv correspondence: matrix-product realization of } \bar{\chi}_{ijkl}^{\gamma J}. \quad (43)$$

$$\bar{\chi}^{(222a),J} = \hat{J}^2 \bar{\eta}^J \bar{\eta}_{occ}^J \bar{\Gamma}^J. \quad (44)$$

The one-body contractions are implemented as the two terms

$$\text{ArXiv correspondence: } f_{ij}^{\text{III}_a} \text{ in } \text{doubleCommutator_1b_Jcoupled_factorized}. \quad (45)$$

$$Z_{pq}^{IIa} = \frac{1}{d_p} \sum_{eJ} \bar{\chi}_{pe,qe}^{(222a),J}, \quad Z_{pq}^{IIc} = -\frac{1}{d_p} \sum_{eJ} \bar{\chi}_{eq,ep}^{(222a),J}, \quad (46)$$

with signs from the stored cross-coupled ordering.

Reference implementation: comm223_232_BruteForce

Source: src/ReferenceImplementations.cc.

This routine builds the two-body part $Z_{pg,qh}^J$ directly. The direct forms below use the permutation symbols that the code expands by explicit direct and exchange loops.

Diagrams Ia, Ib, IVa, IVb

Source: src/ReferenceImplementations.cc.

$$\begin{aligned}
Z_{pg,qh}^{Ia,J_0} &= \frac{1}{2} P_{pg} \sum_{abcdJ_2} \frac{\delta_{jdjp}}{d_p} \hat{J}_2^2 \left(\bar{n}_a \bar{n}_c n_b + \bar{n}_b n_a n_c \right) \eta_{bp,ac}^{J_2} \eta_{ac,bd}^{J_2} \Gamma_{dg,qh}^{J_0}, \\
Z_{pg,qh}^{Ib,J_0} &= \frac{1}{2} P_{qh} \sum_{abcdJ_2} \frac{\delta_{jdjq}}{d_q} \hat{J}_2^2 \left(\bar{n}_a n_b n_c + \bar{n}_b \bar{n}_c n_a \right) \eta_{ad,bc}^{J_2} \eta_{bc,aq}^{J_2} \Gamma_{pg,dh}^{J_0}, \\
Z_{pg,qh}^{IVa,J_0} &= -\frac{1}{2} P_{qh} \sum_{abcdJ_2} \frac{\delta_{jdja}}{d_q} \hat{J}_2^2 \left(\bar{n}_a n_b n_c + \bar{n}_b \bar{n}_c n_a \right) \eta_{bc,aq}^{J_2} \eta_{pg,dh}^{J_0} \Gamma_{ad,bc}^{J_2}, \\
Z_{pg,qh}^{IVb,J_0} &= -\frac{1}{2} P_{pg} \sum_{abcdJ_2} \frac{\delta_{jdjp}}{d_p} \hat{J}_2^2 \left(\bar{n}_a \bar{n}_c n_b + \bar{n}_b n_a n_c \right) \eta_{bp,ac}^{J_2} \eta_{dg,qh}^{J_0} \Gamma_{ac,bd}^{J_2}.
\end{aligned} \tag{47}$$

Diagrams IIa and IIc

Source: src/ReferenceImplementations.cc.

$$\begin{aligned}
Z_{pg,qh}^{IIa,J_0} &= -P_{pg} \sum_{abcdJ_2J_3J_4} \hat{J}_2^2 \hat{J}_3^2 \hat{J}_4^2 \left(\bar{n}_b \bar{n}_d n_c + \bar{n}_c n_b n_d \right) \\
&\quad \times \begin{Bmatrix} j_d & j_g & J_4 \\ j_c & j_b & J_2 \end{Bmatrix} \begin{Bmatrix} j_p & j_a & J_4 \\ j_c & j_b & J_3 \end{Bmatrix} \begin{Bmatrix} j_g & j_p & J_0 \\ j_a & j_d & J_4 \end{Bmatrix} \eta_{cg,db}^{J_2} \eta_{pb,ca}^{J_3} \Gamma_{da,qh}^{J_0}, \\
Z_{pg,qh}^{IIc,J_0} &= -P_{qh} \sum_{abcdJ_2J_3J_4} \hat{J}_2^2 \hat{J}_3^2 \hat{J}_4^2 \left(\bar{n}_b n_c n_d + \bar{n}_c \bar{n}_d n_b \right) \\
&\quad \times \begin{Bmatrix} j_q & j_d & J_4 \\ j_c & j_b & J_2 \end{Bmatrix} \begin{Bmatrix} j_a & j_h & J_4 \\ j_c & j_b & J_3 \end{Bmatrix} \begin{Bmatrix} j_q & j_h & J_0 \\ j_a & j_d & J_4 \end{Bmatrix} \eta_{cd,qb}^{J_2} \eta_{ab,ch}^{J_3} \Gamma_{pg,ad}^{J_0}.
\end{aligned} \tag{48}$$

Diagrams IIb and IId

Source: src/ReferenceImplementations.cc.

$$\begin{aligned}
Z_{pg,qh}^{IIb,J_0} &= -P_{pg}P_{qh} \sum_{abcdJ_2J_3J_4J_5} \hat{J}_2^2 \hat{J}_3^2 \hat{J}_4^2 \hat{J}_5^2 \left(\bar{n}_b n_c n_d + \bar{n}_c \bar{n}_d n_b \right) \\
&\quad \times \left\{ \begin{matrix} j_q & j_d & J_5 \\ j_c & j_b & J_2 \end{matrix} \right\} \left\{ \begin{matrix} j_p & j_a & J_5 \\ j_c & j_b & J_3 \end{matrix} \right\} \left\{ \begin{matrix} J_0 & J_5 & J_4 \\ j_d & j_h & j_q \end{matrix} \right\} \left\{ \begin{matrix} J_0 & J_4 & J_5 \\ j_a & j_p & j_g \end{matrix} \right\} \eta_{dc,bq}^{J_2} \eta_{bp,ac}^{J_3} \Gamma_{ga,hd}^{J_4}, \\
Z_{pg,qh}^{IIc,J_0} &= -P_{pg}P_{qh} \sum_{abcdJ_2J_3J_4J_5} \hat{J}_2^2 \hat{J}_3^2 \hat{J}_4^2 \hat{J}_5^2 \left(\bar{n}_c n_b n_d + \bar{n}_b \bar{n}_d n_c \right) \\
&\quad \times \left\{ \begin{matrix} j_d & j_g & J_5 \\ j_c & j_b & J_2 \end{matrix} \right\} \left\{ \begin{matrix} j_a & j_h & J_5 \\ j_c & j_b & J_3 \end{matrix} \right\} \left\{ \begin{matrix} J_0 & J_5 & J_4 \\ j_d & j_p & j_g \end{matrix} \right\} \left\{ \begin{matrix} J_5 & J_4 & J_0 \\ j_q & j_h & j_a \end{matrix} \right\} \eta_{gc,bd}^{J_2} \eta_{ba,hc}^{J_3} \Gamma_{dp,aq}^{J_4}.
\end{aligned} \tag{49}$$

Diagrams IIe and IIc

Source: src/ReferenceImplementations.cc.

$$\begin{aligned}
Z_{pg,qh}^{IIe,J_0} &= -\frac{1}{2} P_{pg}P_{qh} \sum_{abcdJ_2J_3J_4} \hat{J}_2^2 \hat{J}_3^2 \hat{J}_4^2 \left(\bar{n}_b n_a n_c + \bar{n}_a \bar{n}_c n_b \right) \\
&\quad \times \left\{ \begin{matrix} j_p & j_h & J_4 \\ j_b & j_d & J_2 \end{matrix} \right\} \left\{ \begin{matrix} j_q & j_g & J_4 \\ j_b & j_d & J_3 \end{matrix} \right\} \left\{ \begin{matrix} j_h & j_q & J_0 \\ j_g & j_p & J_4 \end{matrix} \right\} \eta_{ac,bh}^{J_2} \eta_{pd,ac}^{J_2} \Gamma_{bg,qd}^{J_3}, \\
Z_{pg,qh}^{IIc,J_0} &= -\frac{1}{2} P_{pg}P_{qh} \sum_{abcdJ_2J_3J_4} \hat{J}_2^2 \hat{J}_3^2 \hat{J}_4^2 \left(\bar{n}_a \bar{n}_c n_b + \bar{n}_b n_a n_c \right) \\
&\quad \times \left\{ \begin{matrix} j_h & j_p & J_4 \\ j_b & j_d & J_2 \end{matrix} \right\} \left\{ \begin{matrix} j_g & j_q & J_4 \\ j_b & j_d & J_3 \end{matrix} \right\} \left\{ \begin{matrix} j_h & j_q & J_0 \\ j_g & j_p & J_4 \end{matrix} \right\} \eta_{pb,ac}^{J_2} \eta_{ac,dh}^{J_2} \Gamma_{dg,qb}^{J_3}.
\end{aligned} \tag{50}$$

Diagrams IIIa and IIIb

Source: src/ReferenceImplementations.cc.

$$\begin{aligned}
Z_{pg,qh}^{IIIa,J_0} &= P_{pg}P_{qh} \sum_{abcdJ_2J_3J_4J_5} \hat{J}_2^2 \hat{J}_3^2 \hat{J}_4^2 \hat{J}_5^2 \left(\bar{n}_a \bar{n}_c n_b + \bar{n}_b n_a n_c \right) \\
&\quad \times \left\{ \begin{matrix} j_a & j_b & J_5 \\ j_p & j_c & J_2 \end{matrix} \right\} \left\{ \begin{matrix} J_3 & J_0 & J_5 \\ j_p & j_c & j_g \end{matrix} \right\} \left\{ \begin{matrix} j_q & j_d & J_5 \\ j_a & j_b & J_4 \end{matrix} \right\} \left\{ \begin{matrix} J_3 & J_0 & J_5 \\ j_q & j_d & j_h \end{matrix} \right\} \eta_{bp,ca}^{J_2} \eta_{gc,hd}^{J_3} \Gamma_{da,bq}^{J_4}, \\
Z_{pg,qh}^{IIIb,J_0} &= P_{pg}P_{qh} \sum_{abcdJ_2J_3J_4J_5} \hat{J}_2^2 \hat{J}_3^2 \hat{J}_4^2 \hat{J}_5^2 \left(\bar{n}_a n_b n_c + \bar{n}_b \bar{n}_c n_a \right) \\
&\quad \times \left\{ \begin{matrix} j_a & j_b & J_5 \\ j_c & j_q & J_2 \end{matrix} \right\} \left\{ \begin{matrix} J_0 & J_3 & J_5 \\ j_c & j_q & j_h \end{matrix} \right\} \left\{ \begin{matrix} j_d & j_p & J_5 \\ j_a & j_b & J_4 \end{matrix} \right\} \left\{ \begin{matrix} J_0 & J_3 & J_5 \\ j_d & j_p & j_g \end{matrix} \right\} \eta_{cb,aq}^{J_2} \eta_{gd,hc}^{J_3} \Gamma_{ap,db}^{J_4}.
\end{aligned} \tag{51}$$

Diagrams IIIc and IIId

Source: src/ReferenceImplementations.cc.

$$\begin{aligned}
Z_{pg,qh}^{IIIc,J_0} &= -P_{qh} \sum_{abcdJ_2J_3J_4} \hat{J}_2^2 \hat{J}_3^2 \hat{J}_4^2 \left(\bar{n}_a n_b n_c + \bar{n}_b \bar{n}_c n_a \right) \\
&\quad \times \left\{ \begin{matrix} j_q & j_b & J_4 \\ j_c & j_a & J_2 \end{matrix} \right\} \left\{ \begin{matrix} J_3 & J_0 & J_4 \\ j_c & j_a & j_d \end{matrix} \right\} \left\{ \begin{matrix} J_4 & J_3 & J_0 \\ j_h & j_q & j_b \end{matrix} \right\} \eta_{bc,aq}^{J_2} \eta_{pg,cd}^{J_0} \Gamma_{da,hb}^{J_3}, \\
Z_{pg,qh}^{IIId,J_0} &= -P_{pg} \sum_{abcdJ_2J_3J_4} \hat{J}_2^2 \hat{J}_3^2 \hat{J}_4^2 \left(\bar{n}_a \bar{n}_c n_b + \bar{n}_b n_a n_c \right) \\
&\quad \times \left\{ \begin{matrix} j_a & j_p & J_4 \\ j_b & j_c & J_2 \end{matrix} \right\} \left\{ \begin{matrix} J_0 & J_3 & J_4 \\ j_b & j_c & j_d \end{matrix} \right\} \left\{ \begin{matrix} J_4 & J_3 & J_0 \\ j_g & j_p & j_a \end{matrix} \right\} \eta_{bp,ac}^{J_2} \eta_{cd,qh}^{J_0} \Gamma_{ga,db}^{J_3}.
\end{aligned} \tag{52}$$

Diagrams IIIe and IIIf

Source: src/ReferenceImplementations.cc.

$$\begin{aligned}
Z_{pg,qh}^{IIIe,J_0} &= -\frac{1}{2} P_{pg} P_{qh} \sum_{abcdJ_2J_3J_4} \hat{J}_2^2 \hat{J}_3^2 \hat{J}_4^2 \left(\bar{n}_d n_a n_b + \bar{n}_a \bar{n}_b n_d \right) \\
&\quad \times \left\{ \begin{matrix} J_2 & J_3 & J_4 \\ j_p & j_h & j_d \end{matrix} \right\} \left\{ \begin{matrix} J_2 & J_3 & J_4 \\ j_q & j_g & j_c \end{matrix} \right\} \left\{ \begin{matrix} j_h & j_q & J_0 \\ j_g & j_p & J_4 \end{matrix} \right\} \eta_{ab,hd}^{J_2} \eta_{dp,cq}^{J_3} \Gamma_{gc,ab}^{J_2}, \\
Z_{pg,qh}^{III f,J_0} &= -\frac{1}{2} P_{pg} P_{qh} \sum_{abcdJ_2J_3J_4} \hat{J}_2^2 \hat{J}_3^2 \hat{J}_4^2 \left(\bar{n}_c n_a n_b + \bar{n}_a \bar{n}_b n_c \right) \\
&\quad \times \left\{ \begin{matrix} J_2 & J_3 & J_4 \\ j_q & j_g & j_c \end{matrix} \right\} \left\{ \begin{matrix} J_2 & J_3 & J_4 \\ j_p & j_h & j_d \end{matrix} \right\} \left\{ \begin{matrix} j_h & j_q & J_0 \\ j_g & j_p & J_4 \end{matrix} \right\} \eta_{gc,ab}^{J_2} \eta_{dp,cq}^{J_3} \Gamma_{ab,hd}^{J_2}.
\end{aligned} \tag{53}$$

Factorized implementation: comm223_232

Source: src/FactorizedDoubleCommutator.cc.

ArXiv correspondence: this section matches the two-body J-coupled factorized equations in learn/arxiv.txt, namely

$$\Gamma^{IJ}, \Gamma^{IIJ}, \Gamma^{III_a J}, \bar{\bar{\Gamma}}^{III_b J}, \bar{\Gamma}^{III_c J}, \Gamma^{IV_a J}, \bar{\bar{\Gamma}}^{IV_b J}, \bar{\Gamma}^{IV_c J} \tag{54}$$

in Jcoupled_Factorized_DoubleCommutator_twobody, together with the intermediates

$$\chi^\epsilon, \chi^\zeta, \bar{\chi}^\eta, \chi^\theta, \bar{\chi}^\iota, \bar{\bar{\chi}}^\kappa, \chi^\lambda \tag{55}$$

in chi_2b_Jcoupled.

One-body intermediates for Ia, Ib, IVa, IVb

Source: src/FactorizedDoubleCommutator.cc.

ArXiv correspondence tag: this block packages the two one-body-intermediate families

$$\Gamma_{ijkl}^{IJ} \quad \text{and} \quad \Gamma_{ijkl}^{IIJ}, \quad (56)$$

with intermediates

$$\chi_{ij}^\epsilon \quad \text{and} \quad \chi_{ij}^\zeta. \quad (57)$$

In the code-backed form above, the first four insertions are the `\chi~I / \chi~{\epsilon}` family and the second four insertions are the `\chi~{II} / \chi~{\zeta}` family.

The optimized one-body-intermediate path forms two matrices. The code loops over holes i, j for the $\bar{n}_a n_i n_j$ term and over an unrestricted b for the $\bar{n}_a \bar{n}_b n_i$ term:

$$\text{ArXiv correspondence: } \chi_{ij}^\epsilon \text{ and } \chi_{ij}^\zeta \text{ in } \text{chi_2b_Jcoupled}. \quad (58)$$

$$\begin{aligned} \chi_{pq}^I &= \frac{1}{2d_q} \sum_{aijJ_2} \hat{J}_2^2 \bar{n}_a n_i n_j \eta_{ap,ij}^{J_2} \eta_{ij,aq}^{J_2} + \frac{1}{2d_q} \sum_{abiJ_2} \hat{J}_2^2 \bar{n}_a \bar{n}_b n_i \eta_{ip,ab}^{J_2} \eta_{ab,iq}^{J_2}, \\ \chi_{pq}^{II} &= \frac{1}{2d_q} \sum_{aijJ_2} \hat{J}_2^2 \bar{n}_a n_i n_j \Gamma_{ap,ij}^{J_2} \eta_{ij,aq}^{J_2} + \frac{1}{2d_q} \sum_{abiJ_2} \hat{J}_2^2 \bar{n}_a \bar{n}_b n_i \Gamma_{ip,ab}^{J_2} \eta_{ab,iq}^{J_2}. \end{aligned} \quad (59)$$

The resulting two-body update can be written schematically as the ordinary one-body/two-body insertion on the bra and ket sides:

$$\text{ArXiv correspondence: } \Gamma_{ijkl}^{IJ} \text{ and } \Gamma_{ijkl}^{IIJ} \text{ in } \text{Jcoupled_Factorized_DoubleCommutator_twobody}. \quad (60)$$

$$\begin{aligned} Z_{pq,rs}^J &+= \sum_b \left[\chi_{pb}^I \Gamma_{bq,rs}^J + \chi_{qb}^I \Gamma_{pb,rs}^J + \Gamma_{pq,bs}^J \chi_{br}^I + \Gamma_{pq,rb}^J \chi_{bs}^I \right] \\ &+ h_Z \sum_b \left[\chi_{bp}^{II} \eta_{bq,rs}^J + \chi_{bq}^{II} \eta_{pb,rs}^J - \eta_{pq,bs}^J \chi_{br}^{II} - \eta_{pq,rb}^J \chi_{bs}^{II} \right], \end{aligned} \quad (61)$$

with pair-normalization and exchange phases supplied by `GetTBME_norm`.

Two-body intermediates for diagrams II and III

Source: src/FactorizedDoubleCommutator.cc.

ArXiv correspondence tag: this block packages the remaining J-coupled two-body families. The correspondence is grouped by coupling scheme as follows.

- $\chi^{\text{III}}\backslash\Gamma + \text{h_Z}\backslash\Gamma(\chi^{\text{III}})^{\text{T}}$ corresponds to the ordinary-channel χ^{eta} family, i.e.

$$\Gamma_{ijkl}^{\text{III}_a J}. \quad (62)$$

- $\bar{\chi}^{\text{V}}_{\text{RC}}$ and $\chi^{\text{V}}_{\text{final}}=\bar{\chi}^{\text{eta}}, \bar{\chi}^{\text{V}}_{\text{RC}}$ correspond to the double-bar / cross-coupled χ^{eta} family, i.e.

$$\bar{\bar{\Gamma}}_{j\bar{l}k\bar{i}}^{\text{III}_b J}. \quad (63)$$

- $\bar{\chi}_{\text{Gamma}}=\bar{\chi}^{\text{IV}}\backslash\bar{\Gamma}$ corresponds to the barred χ^{theta} family, i.e.

$$\bar{\Gamma}_{i\bar{l}k\bar{j}}^{\text{III}_c J}. \quad (64)$$

- $-\text{eta}\chi^{\text{VI}}-\chi^{\text{VI,II}}\text{eta}$ is the code-packed ordinary-channel realization of the χ^{kappa} family, i.e.

$$\Gamma_{ijkl}^{\text{IV}_a J}. \quad (65)$$

- $\bar{\chi}^{\text{VII}}$ and its inverse-Pandya insertion correspond to the barred/double-barred χ^{iota} and χ^{lambda} families, i.e.

$$\bar{\bar{\Gamma}}_{j\bar{l}k\bar{i}}^{\text{IV}_b J} \quad \text{and} \quad \bar{\Gamma}_{i\bar{l}k\bar{j}}^{\text{IV}_c J}. \quad (66)$$

Manual-check note: the code bundles several arXiv subterms into matrix products, so these tags indicate equation families rather than one printed line per C++ line.

The optimized two-body-intermediate path constructs full cross-coupled matrices $\bar{\eta}$ and $\bar{\Gamma}$ using the Pandya transform

ArXiv correspondence: common Pandya transform for $\bar{\chi}^{\eta}$, $\bar{\chi}^{\iota}$, $\bar{\chi}^{\lambda}$ families. (67)

$$\bar{X}_{ab,cd}^J = \sum_{J'} (-1)^{j_b+j_c+J'} \hat{J}'^2 \begin{Bmatrix} j_a & j_b & J \\ j_c & j_d & J' \end{Bmatrix} X_{ad,bc}^{J'}. \quad (68)$$

The occupation-dressed matrices in the first block are

ArXiv correspondence: occupation factors entering χ^{η} , χ^{ι} , χ^{λ} families. (69)

$$\begin{aligned}
f_{abc} &= \bar{n}_a n_b n_c + n_a \bar{n}_b \bar{n}_c, \\
f_{abd} &= n_a \bar{n}_b n_d + \bar{n}_a n_b \bar{n}_d, \\
f_{bcd} &= n_b n_c \bar{n}_d + \bar{n}_b \bar{n}_c n_d, \\
f_{acd} &= n_a \bar{n}_c n_d + \bar{n}_a n_c \bar{n}_d.
\end{aligned} \tag{70}$$

The code then builds the matrix products

$$\text{ArXiv correspondence: packed intermediates for } \chi^\eta \text{ and } \chi^\kappa \text{ families.} \tag{71}$$

$$\begin{aligned}
\bar{\chi}^{III} &= \bar{\eta} \bar{\eta}_{occ}, \\
\bar{\chi}^V &= \bar{\Gamma} \bar{\eta}_{occ}, \\
\bar{\chi}^{VI} &= \bar{\Gamma} \bar{\eta}_{occ,d}, \\
\bar{\chi}^{VI,II} &= h_\eta \bar{\eta}_{occ,d}^T \bar{\Gamma}.
\end{aligned} \tag{72}$$

After inverse/ordinary channel recoupling, the first direct matrix update is

$$\text{ArXiv correspondence: packed realization of } \Gamma_{ijkl}^{III_a J} \text{ and } \Gamma_{ijkl}^{IV_a J}. \tag{73}$$

$$Z^{IIa+IIc+IIIc+IIId} += \chi^{III} \Gamma + h_Z \Gamma (\chi^{III})^T - \eta \chi^{VI} - \chi^{VI,II} \eta. \tag{74}$$

The IIb/IIId and IIIa/IIIb terms are recoupled through

$$\text{ArXiv correspondence: recoupling step for } \bar{\Gamma}^{III_b J}. \tag{75}$$

$$\bar{X}_{ab,cd}^J = \sum_{J'} (-1)^{j_b+j_c+J'} \hat{J}'^2 \begin{Bmatrix} j_a & j_b & J \\ j_c & j_d & J' \end{Bmatrix} \bar{X}_{ad,bc}^{J'} \tag{76}$$

followed by

$$\text{ArXiv correspondence: matrix-product realization of the } \chi^\eta \text{ cross-coupled family.} \tag{77}$$

$$\chi_{final}^V = \bar{\eta} \bar{\chi}_{RC}^V \tag{78}$$

and the inverse Pandya transform

ArXiv correspondence: inverse transform returning $\bar{\Gamma}^{\equiv \text{III}_b J}$ and related packed families to standard coupling. (79)

$$X_{ij,kl}^J = -(1-P_{ij})(1-P_{kl})(-1)^{J+j_i+j_j} \sum_{J'} (-1)^{J'+j_i+j_k} \hat{J}'^2 \left\{ \begin{matrix} j_j & j_i & J \\ j_k & j_l & J' \end{matrix} \right\} \bar{X}_{jl,ki}^{J'}. \quad (80)$$

For IIe/IIIf the code builds

ArXiv correspondence: packed intermediate for $\bar{\chi}^{\theta J}$. (81)

$$\chi^{IV} = \eta \eta_{occ,c} + (\eta \eta_{occ,d})^T, \quad (82)$$

and for IIIe/IIIIf it builds

ArXiv correspondence: packed intermediate for $\bar{\chi}^{\iota J}$ and $\chi^{\lambda J}$ families. (83)

$$\chi^{VII} = \Gamma \eta_{occ,d} + h_\eta \eta_{occ,d}^T \Gamma. \quad (84)$$

These are recoupled to $\bar{\chi}^{IV}$ and $\bar{\chi}^{VII}$, then IIe/IIIf use

ArXiv correspondence: matrix-product realization of $\bar{\Gamma}_{i\bar{l}k\bar{j}}^{\text{III}_c J}$. (85)

$$\bar{\chi}_\Gamma = \bar{\chi}^{IV} \bar{\Gamma}, \quad (86)$$

with the same inverse Pandya transform. The comments in the code write these final two as

ArXiv correspondence: packed inverse-Pandya forms for $\bar{\Gamma}^{\text{III}_c J}$ and $\bar{\Gamma}^{\text{IV}_c J}$. (87)

$$\begin{aligned} Z_{ij,kl}^{IIe,J} &= -\frac{1}{2}(1-P_{ij})(1-P_{kl}) \sum_{J'} \hat{J}'^2 \left\{ \begin{matrix} j_i & j_j & J \\ j_k & j_l & J' \end{matrix} \right\} \bar{\chi}_{\Gamma;il,kj}^{J'}, \\ Z_{ij,kl}^{IIIf,J} &= -\frac{1}{2}(1-P_{ij})(1-P_{kl}) \sum_{J'} \hat{J}'^2 \left\{ \begin{matrix} j_i & j_j & J \\ j_k & j_l & J' \end{matrix} \right\} \bar{\chi}_{\Gamma,II;il,kj}^{J'}. \end{aligned} \quad (88)$$

Reference and factorized implementation: `comm223_132`

Sources: `src/ReferenceImplementations.cc`, `src/FactorizedDoubleCommutator.cc`.

The `223_132` contribution is the remaining two-body part of

$$[\eta_2, [\eta_2, \Gamma_2]_{3b}]_{2b} \quad (89)$$

where one of the contractions is represented by the one-body part of `Eta`. In the reference code it is implemented by `comm223_132_impl` with three independently callable pieces:

$$Z^{223_132} = Z^{\text{ladder}} + Z^{\text{cross}} + Z^{\text{onebody}}. \quad (90)$$

The wrappers `comm223_132_ladder`, `comm223_132_cross`, and `comm223_132_onebody` select these pieces separately.

ArXiv correspondence tag: `223_132` is not covered by the scalar factorized equations in `learn/arxiv.txt`. Manual checking for this section should instead use the dedicated AMC-derived note and the code-backed equations in this file.

Common one-body contraction

Source: `src/ReferenceImplementations.cc`, `src/FactorizedDoubleCommutator.cc`.

All three pieces use the occupation-weighted one-body contraction

$$f_{ab} \equiv (\bar{n}_a n_b - n_a \bar{n}_b) \eta_{ba}, \quad j_a = j_b, \quad (91)$$

or the transposed matrix element η_{ab} depending on the orientation of the diagram. The code enforces the angular-momentum delta by looping over `GetOneBodyChannel(oa.l, oa.j2, oa.tz2)`.

Ladder piece

Source: `src/ReferenceImplementations.cc`, `src/FactorizedDoubleCommutator.cc`.

For a two-body channel with $J_0 = J_1 = J$, the direct ladder contribution is

$$Z_{ij,kl}^{\text{ladder},J} = \sum_{abc} f_{ab} \left(\eta_{ca,kl}^J \Gamma_{ij,cb}^J - \Gamma_{ca,kl}^J \eta_{ij,cb}^J \right). \quad (92)$$

The C++ loops also include the exchanged storage of the intermediate ket pair (c, a) and apply the usual $1/\sqrt{2}$ pair normalization for $i = j$ or $k = l$ before storing the TBME.

Direct one-body-contraction piece

Source: src/ReferenceImplementations.cc, src/FactorizedDoubleCommutator.cc.

The reference implementation forms two one-body intermediates

$$\begin{aligned}\chi_{pq}^\eta &= \frac{1}{d_q} \sum_{ijJ_2} \hat{J}_2^2 (\bar{n}_i n_j - n_i \bar{n}_j) \eta_{pi,qj}^{J_2} \eta_{ji}, \\ \chi_{pq}^\Gamma &= \frac{1}{d_q} \sum_{ijJ_2} \hat{J}_2^2 (\bar{n}_i n_j - n_i \bar{n}_j) \Gamma_{pi,qj}^{J_2} \eta_{ji}.\end{aligned}\tag{93}$$

These are then inserted into the external two-body line as

$$\begin{aligned}Z_{ij,kl}^{\text{onebody},J} &= \sum_a \left(\chi_{ia}^\eta \Gamma_{aj,kl}^J + \chi_{ja}^\eta \Gamma_{ia,kl}^J - \Gamma_{ij,al}^J \chi_{ak}^\eta - \Gamma_{ij,ka}^J \chi_{al}^\eta \right) \\ &\quad - \sum_a \left(\chi_{ia}^\Gamma \eta_{aj,kl}^J + \chi_{ja}^\Gamma \eta_{ia,kl}^J - \eta_{ij,al}^J \chi_{ak}^\Gamma - \eta_{ij,ka}^J \chi_{al}^\Gamma \right),\end{aligned}\tag{94}$$

with the exchange phases supplied explicitly in the reference routine as

$$(-1)^{J+j_i+j_j}, \quad (-1)^{J+j_k+j_l}\tag{95}$$

when the insertion is on the exchanged bra or ket leg. Written with those phases displayed, the eight explicit reference-code terms are

$$\begin{aligned}Z_{ij,kl}^{\text{onebody},J} &= \sum_{abcJ_2} \hat{J}_2^2 f_{ab} \left[\frac{1}{d_i} \eta_{ia,cb}^{J_2} \Gamma_{cj,kl}^J - \frac{(-1)^{J+j_i+j_j}}{d_j} \eta_{ja,cb}^{J_2} \Gamma_{ci,kl}^J \right. \\ &\quad - \frac{1}{d_k} \Gamma_{ij,cl}^J \eta_{ac,bk}^{J_2} + \frac{(-1)^{J+j_k+j_l}}{d_l} \Gamma_{ij,ck}^J \eta_{ac,bl}^{J_2} \\ &\quad + \frac{1}{d_l} \eta_{ij,kc}^J \Gamma_{ca,lb}^{J_2} - \frac{(-1)^{J+j_k+j_l}}{d_k} \eta_{ij,lc}^J \Gamma_{ca,kb}^{J_2} \\ &\quad \left. - \frac{1}{d_j} \Gamma_{ja,cb}^{J_2} \eta_{ic,kl}^J + \frac{(-1)^{J+j_i+j_j}}{d_i} \Gamma_{ia,cb}^{J_2} \eta_{jc,kl}^J \right].\end{aligned}\tag{96}$$

Cross-coupled reference piece

Source: src/ReferenceImplementations.cc.

The direct reference implementation writes the cross term in the normal-coupled basis with a nine-j recoupling. A representative direct term is

$$Z_{ij,kl}^{\text{cross},J_0} \supset \sum_{abcJ_2J_3} (-1)^{J_2+J_3+j_a+j_c+J_0} \hat{J}_2^2 \hat{J}_3^2 \begin{Bmatrix} j_i & j_c & J_2 \\ j_j & J_3 & j_a \\ J_0 & j_l & j_k \end{Bmatrix} \eta_{ic,ka}^{J_2} \eta_{ab} \Gamma_{bj,cl}^{J_3} (\bar{n}_a n_b - n_a \bar{n}_b). \quad (97)$$

The code expands this seed into the eight antisymmetrized variants generated by exchanging $i \leftrightarrow j$ and $k \leftrightarrow l$, with signs and extra phases

$$1, \quad (-1)^{j_i+j_j}, \quad (-1)^{j_k+j_l}, \quad (-1)^{J_0+j_i+j_j+j_k+j_l}. \quad (98)$$

The alternate direct term in the lambda is the same recoupling with the one-body contraction placed on the other side of the two-body matrix element,

$$-\eta_{ib,kc}^{J_2} \eta_{ab} \Gamma_{cj,al}^{J_3}, \quad (99)$$

and its exchanged partners.

Factorized cross implementation

Source: `src/FactorizedDoubleCommutator.cc`.

The optimized `FactorizedDoubleCommutator::comm223_132_cross` evaluates the same cross piece through cross-coupled matrix products. It first forms

$$\bar{X}_{ab,cd}^J = \sum_{J'} (-1)^{j_b+j_c+J'} \hat{J}'^2 \begin{Bmatrix} j_a & j_b & J \\ j_c & j_d & J' \end{Bmatrix} X_{ad,bc}^{J'}, \quad X \in \{\eta, \Gamma\}. \quad (100)$$

The one-body-contraction matrix in the same cross-coupled space is

$$\bar{\eta}_{ab,cd}^{(1),J} = \delta_{ac} \eta_{db} (\bar{n}_b n_d - n_b \bar{n}_d) + \delta_{bd} \eta_{ac} (\bar{n}_a n_c - n_a \bar{n}_c). \quad (101)$$

The cross intermediate is the simple matrix product

$$\bar{\chi}^{132,J} = \bar{\eta}^J \bar{\eta}^{(1),J} \bar{\Gamma}^J. \quad (102)$$

It is transformed back to the normal-coupled TBME through the same inverse Pandya antisymmetrizer used elsewhere in the factorized double commutator:

$$X_{ij,kl}^J = -(1-P_{ij})(1-P_{kl})(-1)^{J+j_i+j_j} \sum_{J'} (-1)^{J'+j_i+j_k} \hat{J}'^2 \begin{Bmatrix} j_j & j_i & J \\ j_k & j_l & J' \end{Bmatrix} \bar{X}_{jl,ki}^{J'}. \quad (103)$$

In component form the code accumulates four inverse-Pandya pieces,

$$Z_{ij,kl}^{\text{cross},J} = C_{ij,kl} - (-1)^{j_i+j_j-J} C_{ji,kl} - (-1)^{j_k+j_l-J} C_{ij,lk} + (-1)^{j_i+j_j+j_k+j_l} C_{ji,lk}, \quad (104)$$

where each C is a J' sum over the appropriate element of $\bar{\chi}^{132,J'}$.